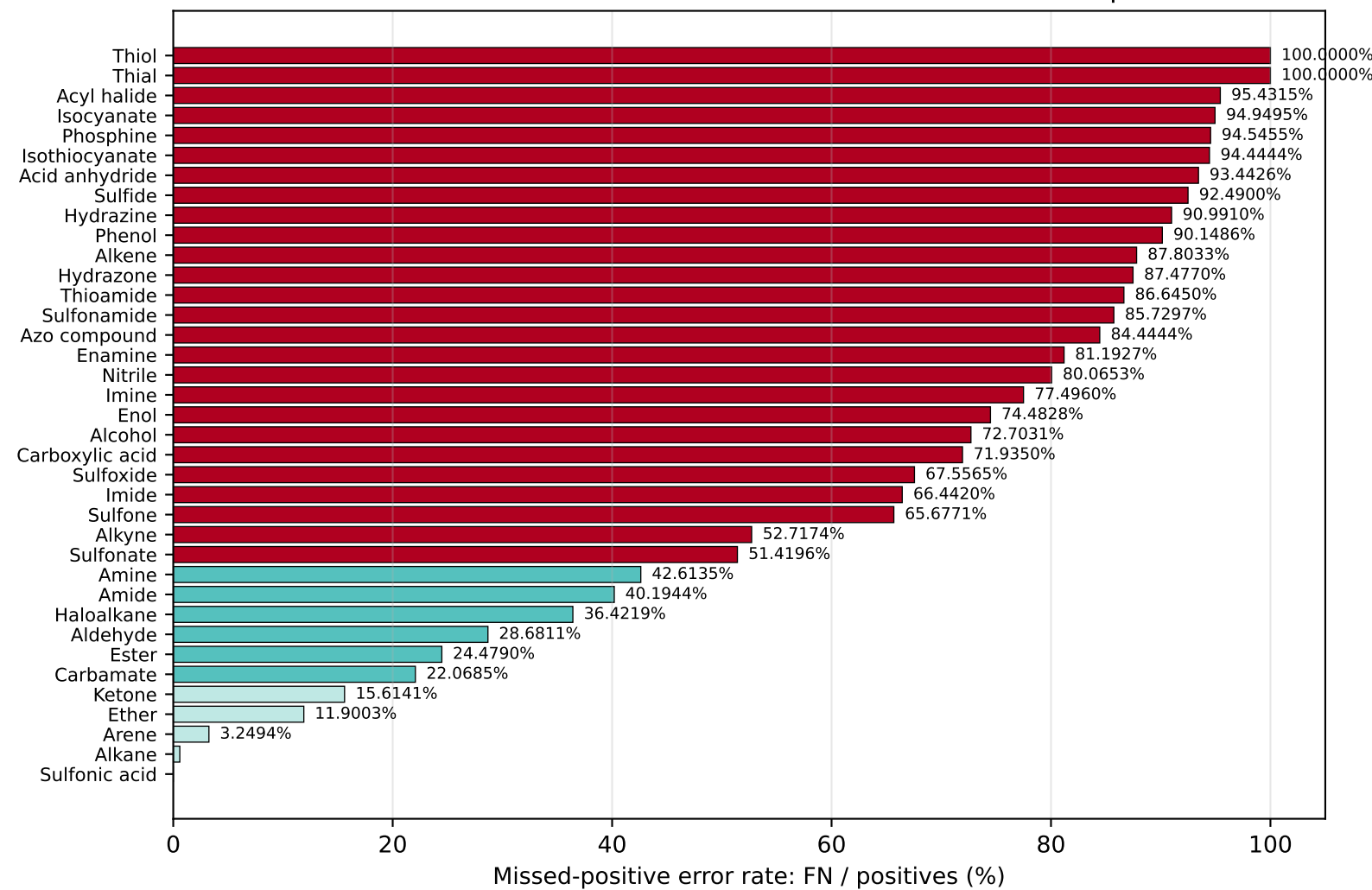
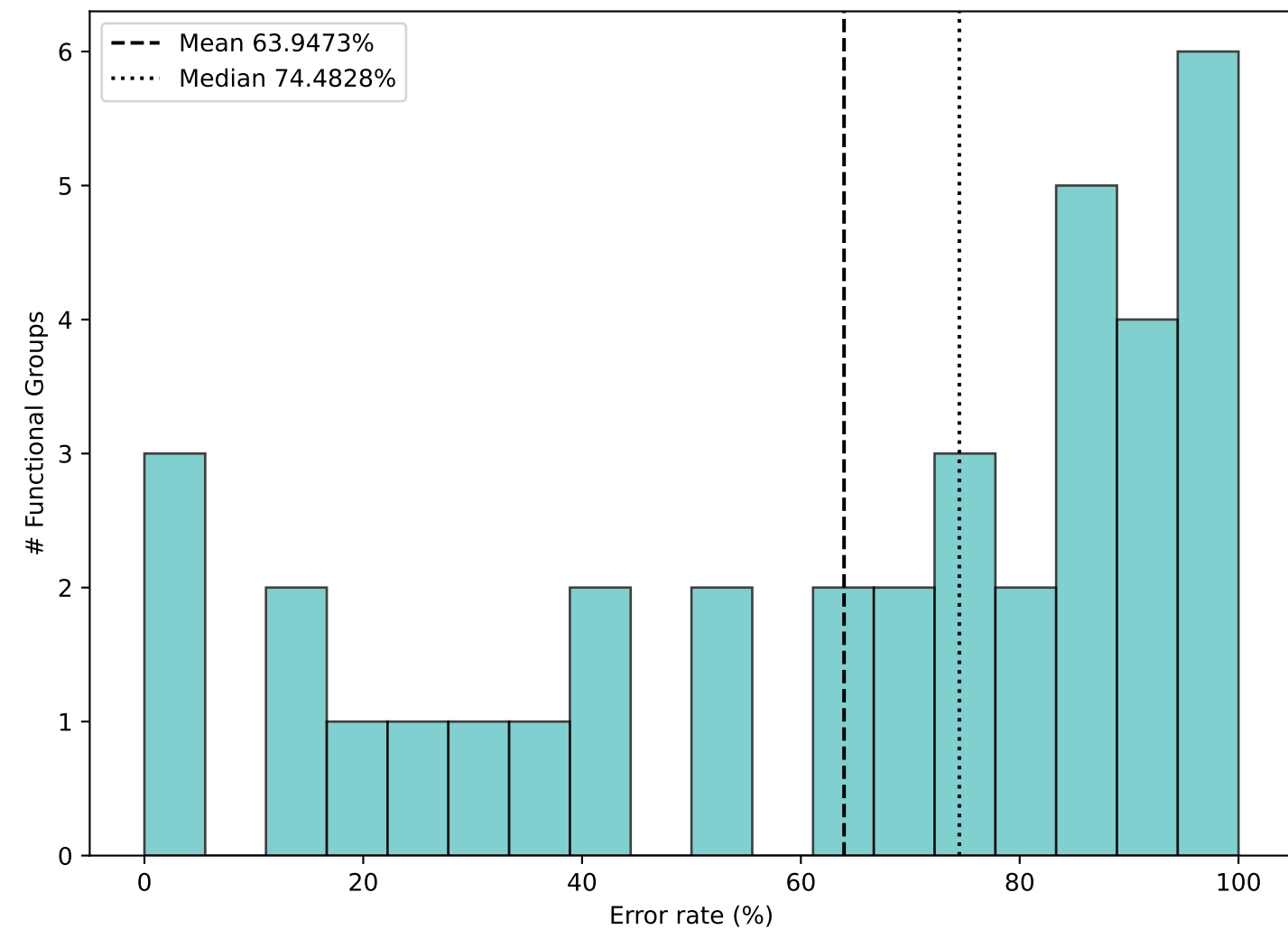


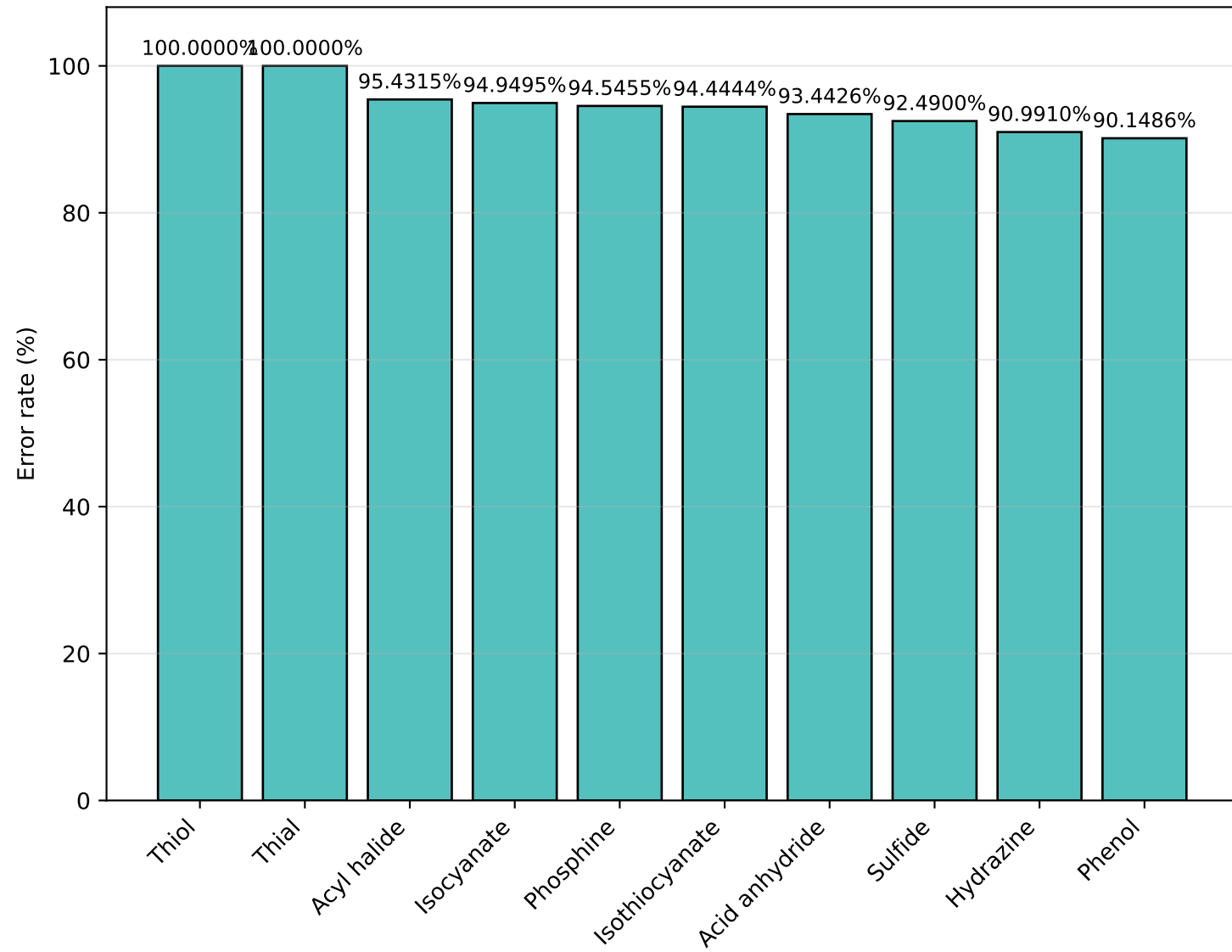
XGBoost C-NMR Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



XGBOOST C-NMR VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro:	0.8014
F1 macro:	0.4687
Precision micro:	0.8840
Recall micro:	0.7329
Hamming accuracy:	0.9531

Error-rate statistics:

Mean:	63.9473%
Median:	74.4828%
Std:	31.8482%
Max:	100.0000% (Thiol)
Min:	0.0000% (Sulfonic acid)

Problem classes:

Severe (>50%):	26 groups
High (20-50%):	6 groups
Moderate (5-20%):	2 groups
Low (<=5%):	3 groups